

Program Increment (PI) 1

Goal

Deliver a **clickable UI prototype** that supports the main MVP user scenarios and demonstrates a clear user experience flow.

Deliverables

Project Pitch & Project Demo #1

PI Planning

Date: 01/27

Purpose

- Define MVP scope
 -  CoursePilot Potential Features
- Create epics in [Jira Board](#)
- Write and refine user stories
 - Add to Jira under appropriate epic
- Estimate story points
- Assign stories to Sprint 1 and potentially 2

Sprint 1

Dates: 01/28 – 02/10

Focus: Foundations & Architecture

- Create wireframes in Figma
- Develop visual style guide
- Refine project pitch
- Create UML diagrams (system overview, key components)

Epic 1: Infrastructure Initialization

Goal: Initialize the development environments so the team can start coding in Sprint 2.

Story 1.1: Frontend Repository Setup

- **Description:** Initialize the Next.js application structure and CI/CD pipeline.
- **Task 1.1.A:** Initialize Git repository with `create-next-app` (TypeScript, Tailwind CSS, ESLint).
 - **Assignee:** Daniel Amoako-Frimpong
 - **Deadline:** Feb 5 (Thursday)
- **Task 1.1.B:** Configure Vercel project and link to GitHub repository for automatic deployments on push.
 - **Assignee:** Daniel Amoako-Frimpong
 - **Deadline:** Feb 5 (Thursday)

- **Task 1.1.C:** Install and configure the Firebase Client SDK (`firebase/app`, `firebase/auth`, `firebase/firestore`).
 - **Assignee:** Daniel Amoako-Frimpong
 - **Deadline:** Feb 7 (Saturday)

Story 1.2: Backend API Setup

- **Description:** Initialize the Python FastAPI service and cloud hosting.
- **Task 1.2.A:** Initialize Git repository with Python virtual environment and `requirements.txt` (FastAPI, Uvicorn).
 - **Assignee:** Evan Tomaine
 - **Deadline:** Feb 5 (Thursday)
- **Task 1.2.C:** Deploy the "Hello World" service to Render/Railway to verify the CI/CD pipeline is working.
 - **Assignee:** Evan Tomaine
 - **Deadline:** Feb 8 (Sunday)

Story 1.3: Firebase Project Configuration

- **Description:** Set up the cloud backend services (Auth, DB, Storage).
- **Task 1.3.A:** Create new Firebase Project "CoursePilot-Dev" in the console.
 - **Assignee:** Matt Zandi
 - **Deadline:** Feb 5 (Thursday)
- **Task 1.3.B:** Enable Authentication providers (Google Sign-In and Email/Password).
 - **Assignee:** Matt Zandi
 - **Deadline:** Feb 5 (Thursday)
- **Task 1.3.C:** Provision Firestore Database in `nam5` (us-central) and set up initial Security Rules (allow read/write for auth users).
 - **Assignee:** Matt Zandi
 - **Deadline:** Feb 6 (Friday)

Epic 2: System Architecture & Design

Goal: Define the data structures and API contracts before coding begins.

Story 2.1: Database Schema Design

- **Description:** Define how data will be structured in NoSQL Firestore.
- **Task 2.1.A:** Draft JSON document structure for the `Users` collection (profile, preferences).
 - **Assignee:** Vaidik Vadhavana
 - **Deadline:** Feb 6 (Friday)
- **Task 2.1.B:** Draft JSON document structure for the `Syllabi` and `Events` collections.
 - **Assignee:** Vaidik Vadhavana
 - **Deadline:** Feb 6 (Friday)
- **Task 2.1.C:** Review schema with Frontend team to ensure it meets UI needs.
 - **Assignee:** Vaidik Vadhavana

- **Deadline:** Feb 10 (Tuesday)

Story 2.2: System Documentation

- **Description:** Create the blueprints for the system.
- **Task 2.2.A:** Create High-Level Architecture Diagram (UML) showing Microservices flow.
 - **Assignee:** Weston Trautmann
 - **Deadline:** Feb 8 (Sunday)
- **Task 2.2.B:** Create Sequence Diagram for the "Syllabus Upload -> Parsing -> DB" flow.
 - **Assignee:** Weston Trautmann
 - **Deadline:** Feb 8 (Sunday)
- **Task 2.2.C:** Write API Specification (OpenAPI/Swagger) defining the `/upload` and `/events` endpoints.
 - **Assignee:** Michael Nguyen
 - **Deadline:** Feb 9 (Monday)

Epic 3: UX/UI Design

Goal: Deliver the visual guide for the frontend developers.

Story 3.1: Wireframing & Prototyping

- **Description:** Design the look and feel of the application.
- **Task 3.1.A:** Create Low-Fidelity Wireframes for the Login Page and Dashboard Home.
 - **Assignee:** Nicolette Fulcomer
 - **Deadline:** Feb 6 (Friday)
- **Task 3.1.B:** Create Low-Fidelity Wireframes for the Calendar View and Syllabus Upload Modal.
 - **Assignee:** Lauryn Do-Diallo
 - **Deadline:** Feb 6 (Friday)
- **Task 3.1.C:** Create High-Fidelity clickable prototype in Figma linking these screens.
 - **Assignee:** Nicolette Fulcomer
 - **Deadline:** Feb 10 (Tuesday)

Story 3.2: Visual Style Guide

- **Description:** Establish the visual language.
- **Task 3.2.A:** Define Color Palette (Primary, Secondary, Alerts) and Typography.
 - **Assignee:** Lauryn Do-Diallo
 - **Deadline:** Feb 8 (Sunday)
- **Task 3.2.B:** Create component library in Figma (Buttons, Cards, Inputs).
 - **Assignee:** Lauryn Do-Diallo
 - **Deadline:** Feb 10 (Tuesday)

Epic 4: AI Pipeline Research (Spike)

Goal: Prove the AI technology stack works before implementation.

Story 4.1: PDF Parsing Feasibility

- **Description:** Test LlamaParse against real syllabi.
- **Task 4.1.A:** Gather 5 sample syllabi with varying formats (tables, columns, bad scanning).
 - **Assignee:** Joshua Oti
 - **Deadline:** Feb 5 (Thursday)
- **Task 4.1.B:** Run samples through LlamaParse API and document the quality of text extraction.
 - **Assignee:** Joshua Oti
 - **Deadline:** Feb 7 (Saturday)

Story 4.2: LLM Prompt Engineering

- **Description:** Create the system prompt that converts text to JSON.
- **Task 4.2.A:** Draft the system prompt for GPT-4o-mini (e.g., "You are an assistant... output only JSON...").
 - **Assignee:** Pradeep Mathiyarasu
 - **Deadline:** Feb 6 (Friday)
- **Task 4.2.B:** Test the prompt with the extracted text from Task 4.1.B and verify JSON validity.
 - **Assignee:** Pradeep Mathiyarasu
 - **Deadline:** Feb 8 (Sunday)

Epic 5: Project Management

Goal: Meet course administrative requirements.

Story 5.1: Agile Setup

- **Description:** Configure Jira and documentation.
- **Task 5.1.A:** Create Jira Project and set up Board columns (To Do, In Progress, Review, Done).
 - **Assignee:** Michael Nguyen
 - **Deadline:** Feb 4 (Wednesday)
- **Task 5.1.B:** Input all User Stories and Tasks into Jira and assign Story Points
 - **Assignee:** Michael Nguyen & Lauryn Do-Diallo
 - **Deadline:** Feb 5 (Thursday)

Story 5.2: Course Deliverables

- **Description:** Prepare for class presentations.
- **Task 5.2.A:** Finalize "Project Pitch Deck" slides.
 - **Assignee:** Michael Nguyen & Evan Tomaine
 - **Deadline:** Feb 2 (Completed/Past Due)

→ Supports: **Project Pitch (2/02)**

Sprint 2

Dates: 02/11 – 02/24

Focus: Clickable MVP and Data Flow Simulation

→ Supports: **Demo #1 (2/23)**

Program Increment (PI) 2

Goal

<Add goal>

Planning Date

02/24

Purpose

- Sprint 2 retrospective
- Re-estimate backlog
- Plan implementation-focused sprints